

Ecological bias distorts effect sizes in federated COVID-19 analysis: evidence from 42 million patient records

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Abstract

Background: Multi-site medical studies and federated analysis networks routinely pool site-level summaries to draw conclusions about individual patients. The ecological fallacy—inferring individual-level effects from group-level data—is well known but rarely quantified in large-scale clinical datasets.

Methods: We compared ecological (site-level) regression with individual-level logistic regression on 42 million COVID-19 patient records from the United States (38 million CDC records, 7 racial/ethnic pseudo-sites) and Mexico (4 million confirmed cases, 32 states). We applied meta-analytic methods and structural diagnostics to the 4CE consortium’s neurological COVID-19 dataset (22,000 patients, 21 hospitals, 6 countries).

Results: In both datasets, ecological regression detects a significant relationship between age and mortality, but the ecological coefficient (a linear slope) is incommensurable with the individual-level truth (a multiplicative odds ratio). CDC: ecological $\beta = +0.55$ ($p = 0.02$) vs. individual OR = 9.9 ($p < 0.001$). Mexico: ecological $\beta = +1.31$ ($p < 0.0001$) vs. individual OR = 11.3 ($p < 0.001$). In the 4CE dataset, meta-analytic methods diverge by 350-fold on the pooled effect ($I^2 = 99.8\%$), elderly proportion explains most between-site variation ($\beta = +14.2$, $p < 10^{-15}$), and a multivariate consistency test detects coordinated heterogeneity ($z = 25$, $p < 0.0001$) invisible to standard tests.

Conclusions: Ecological regression produces coefficients that are incommensurable with patient-level effects—a different quantity, not a noisy estimate. Pooled site-level summaries should be treated as hypotheses for individual-level testing, not as established findings.

Keywords: ecological fallacy, federated analysis, meta-analysis, ecological bias, COVID-19, multi-site studies, aggregation bias

1 Introduction

Suppose ten hospitals each report the fraction of their COVID-19 patients who died. A meta-analysis averages those fractions and reports the result as “the” mortality rate. The same logic underlies federated analysis networks such as ENACT, TriNetX, and PCORnet, where hospitals share summary statistics rather than individual records. This approach works when the hospitals are measuring the same thing in the same kind of patient.

They rarely are.

Hospitals differ in who walks through the door. One serves a young urban population; another, a rural elderly community. One codes aggressively for neurological complications; another barely records them. The average across these hospitals is a real number, but it may not describe any actual hospital or any actual patient. Worse, it can point in the wrong direction entirely.

This problem—drawing conclusions about individuals from group-level data—is the *ecological fallacy* Robinson [1950]. The structural mechanisms behind it—aggregation bias, cross-level confounding, and specification error—are well understood Greenland and Morgenstern [1992], Greenland [2001], and formal frameworks for ecological inference exist King [1997], Wakefield [2003, 2008]. Known since 1950 and taught in every epidemiology course, it is routinely acknowledged and routinely ignored. Multi-site consortium studies and federated analysis networks typically have access only to site-level summaries and treat them as proxies for patient-level truth.

We test how badly this matters using 42 million individual COVID-19 patient records from two countries, plus a third dataset of 22,000 patients from 21 hospitals across 6 countries. The CDC dataset uses race/ethnicity categories as pseudo-sites (7 groups, excluding categories with ambiguous demographic composition). Mexico’s 32 states provide a confirmatory test on real geographic sites. In both cases, ecological regression detects a statistically significant relationship between age and mortality—but the ecological coefficient is a linear slope, while the individual-level truth is a multiplicative odds ratio an order of magnitude larger:

- **CDC (38 million records, 7 pseudo-sites):** ecological $\beta = +0.55$, $p = 0.02$; individual OR = 9.9, $p < 0.001$.
- **Mexico (4 million records, 32 states):** ecological $\beta = +1.31$, $p < 0.0001$; individual OR = 11.3, $p < 0.001$.

The ecological regression “works” in both cases—it finds a significant positive relationship—but the number it reports is meaningless as an estimate of the patient-level effect.

We then ask: if federated networks are constrained to site-level summaries, can anything be done? Structural consistency tests—methods that examine the *structure* of relationships across sites, rather than averaging over them—can diagnose when site-level pooling is unreliable. Applied to the 4CE consortium’s neurological COVID-19 dataset, these methods detect coordinated heterogeneity invisible to standard meta-analysis. Federated analyses should treat their pooled summaries as hypotheses for individual-level testing.

2 Ecological Bias in Two Datasets

2.1 CDC, United States: ecological distortion with pseudo-sites

We start with the simplest possible question: does age predict COVID-19 mortality? The answer is obviously yes. The question is whether site-level analysis recovers the right answer.

The CDC Case Surveillance dataset contains 38.2 million confirmed COVID-19 cases with age group, sex, race/ethnicity, hospitalization status, ICU admission, and death. We used 7 race/ethnicity categories as pseudo-sites (excluding “Unknown” and “Missing” categories with ambiguous demographic composition; groups range from $n = 13,548$ to $n = 6,165,036$), mimicking what a consortium study would have if each racial/ethnic group were a separate hospital reporting summary statistics. This is a deliberate simplification that maximizes compositional confounding; the specific confounders differ from those in hospital-based consortia (see Limitations).

Site-level analysis aggregated each group’s mortality rate (deaths/cases) and proportion of patients aged 80+, then regressed one on the other across the 7 groups. Result: a significant positive relationship ($\beta = +0.55$ mortality-rate units per unit proportion elderly, $p = 0.02$, $R^2 = 0.69$). The ecological regression detects that age predicts mortality.

Individual-level analysis on the same 38.2 million patients found that being over 80 multiplies the odds of death by 9.9 ($p < 0.001$). After adjusting for sex and comorbidities, patients gain 2.4

times the odds of dying for each additional decade of age ($p < 0.001$). Males have 1.7 times the odds; patients with underlying conditions have 1.5 times the odds.

The ecological regression gets the direction right, but the quantities are incommensurable (Figure 1A): a linear slope of +0.55 stands in for a multiplicative odds ratio of 9.9. The ecological coefficient is not a noisy estimate of the individual-level effect—it is a different quantity entirely.

Three pre-registered supplementary analyses illuminate the role of the grouping strategy. When all 105 million CDC records (including probable cases and all race/ethnicity categories, compared to the 38.2 million confirmed cases used in the primary analysis) are randomly partitioned into 9 groups (destroying the race/ethnicity structure), ecological regression achieves only 4.8% power (95% Wilson CI [3.2%, 7.0%])—indistinguishable from the false-positive rate (Supplement S6; Figure 2). The race/ethnicity grouping succeeds because different racial/ethnic groups have different age distributions; random grouping averages out these differences, leaving no signal for ecological regression to detect. Duncan–Davis ecological inference bounds on the elderly mortality rate span [0, 24.4%], containing the true 9.91% rate but too wide to be informative (Supplement S4).

2.2 Mexico: ecological distortion with real geographic sites

Mexico’s COVID-19 surveillance data provides a confirmatory test on real geographic sites: 3,993,464 confirmed cases across 32 real geographic states, each corresponding to the state of the treating medical unit.

Site-level analysis across the 32 states finds a significant relationship between the proportion of patients aged 70+ and the state mortality rate ($\beta = +1.31$ mortality-rate units per unit proportion elderly, $p < 0.0001$, $R^2 = 0.50$). Concretely, a 10-percentage-point increase in a state’s elderly share predicts a 13.1-percentage-point increase in state mortality. The ecological regression “works”—it detects a real signal, as does the CDC analysis.

Individual-level analysis on the same 4 million patients reveals the true effect: patients aged 70 or older have 11.3 times the odds of dying ($p < 0.001$). After adjusting for sex and 9 individual comorbidities (diabetes, hypertension, obesity, cardiovascular disease, chronic kidney disease, COPD, asthma, immunosuppression, and smoking), the adjusted odds ratio remains 8.0 ($p < 0.001$). Males have 1.7 times the odds; patients with any comorbidity have 3.8 times the odds.

Site-level analysis gets the direction right and the size wrong (Figure 1B): a linear slope of +1.31 standing in for an odds ratio of 11. The two numbers don’t even measure the same kind of thing.

Supplementary analyses quantify this distortion directly. To isolate the aggregation bias, we fitted the individual-level logistic model (elderly, sex, comorbidity) and computed the ecological slope that *would* arise if the ecological regression faithfully summarized the individual-level relationship. The observed ecological slope is 2.09 times the expected slope—a 109% amplification from aggregation bias (Supplement S3). Two individual-level pooling methods—two-stage IPD meta-analysis (per-state logistic regression pooled via DerSimonian–Laird random effects) and one-stage IPD (logistic with state fixed effects)—recover odds ratios of 5.34 and 5.22, respectively, agreeing within 2% (Supplement S9). The ecological regression produces a meaningless slope; both IPD methods recover the individual-level effect.

These results are complementary. Both datasets show ecological regression detecting a statistically significant relationship, but in both cases the ecological coefficient is a linear slope where the patient-level truth is a multiplicative odds ratio an order of magnitude larger—a different quantity, not a noisy estimate of the same one. Any federated analysis relying on site-level summaries faces this distortion.

Table 1: Ecological distortion in both datasets. Site-level regression detects a significant relationship in both cases, but the ecological coefficient (a linear slope) is incommensurable with the individual-level truth (a multiplicative odds ratio).

Dataset	Analysis level	Effect size	<i>p</i> -value
<i>CDC (38.2M cases, 7 pseudo-sites):</i>			
	Site-level regression	$\beta = +0.55$	0.02
	Individual: age per decade	OR = 2.40	< 0.001
	Individual: age 80+	OR = 9.88	< 0.001
<i>Mexico (4.0M cases, 32 states):</i>			
	Site-level regression	$\beta = +1.31$	< 0.0001
	Individual: age 70+	OR = 11.26	< 0.001
	Individual: adjusted	OR = 7.97	< 0.001

2.3 Why does this happen?

The ecological fallacy arises from two mechanisms operating simultaneously: cross-level confounding and aggregation bias.

Cross-level confounding. The 7 CDC pseudo-sites differ in sex ratios (15% to 65% male) and comorbidity prevalence (0.5% to 14%). At the individual level, confounders are controlled directly: the logistic regression asks “among patients with the same sex and comorbidity status, does age predict death?” At the ecological level, these confounders are entangled with the grouping variable. The ecological regression detects the age–mortality relationship ($p = 0.02$) because the 7 groups have sufficiently distinct age distributions, but the slope it estimates reflects the joint variation of age, sex, and comorbidity across groups, not the isolated effect of age.

Aggregation bias. Even without confounding, the ecological regression fits a linear slope to group-level proportions, while the true patient-level relationship is a nonlinear odds ratio. Mexico illustrates this directly: the 32 states have less extreme compositional variation than the CDC pseudo-sites, and the ecological regression detects a strong directional signal ($p < 0.0001$), but the ecological slope of +1.31 is incommensurable with the individual-level odds ratio of 11.3. The direction is preserved; the magnitude and functional form are lost.

A simulation exercise (Supplement S1) isolates the role of grouping strategy. We generated individual-level data within each CDC site using the known odds ratios (age OR = 9.88, comorbidity OR = 1.5, sex OR = 1.7) and exact site compositions. Ecological regression on the simulated site-level data achieves 100% power—but only when grouping preserves demographic differences between sites. Random grouping of the same 105 million records into 9 groups achieves 4.8% power (Figure 2), because random assignment averages out the age-distribution differences that the ecological regression depends on.

Noncollapsibility compounds the distortion. The odds ratio is noncollapsible: even without confounding, a marginal (population-averaged) odds ratio differs from the conditional (stratum-specific) odds ratio whenever the outcome is common or the exposure effect varies across strata Greenland et al. [1999]. Aggregation across sites collapses strata, so the ecological analysis faces noncollapsibility on top of cross-level confounding and specification error. This means the ecological

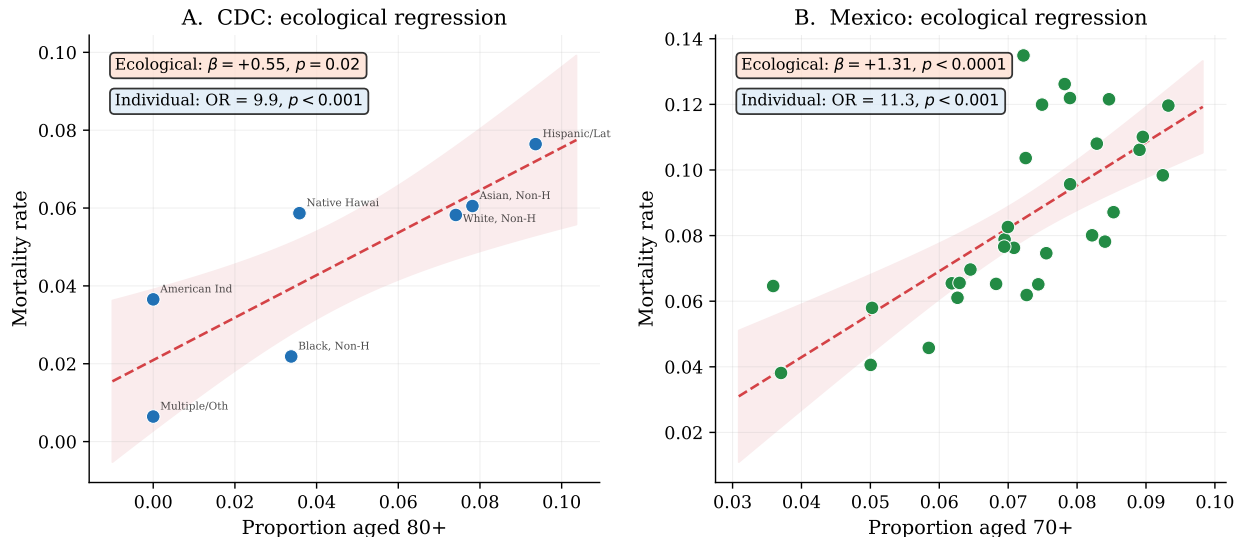


Figure 1: Site-level ecological regressions for both datasets. **(A)** CDC: 7 race/ethnicity pseudo-sites show a significant relationship ($\beta = +0.55, p = 0.02$), but this linear slope is incommensurable with the individual-level odds ratio of 9.9. **(B)** Mexico: 32 states show a stronger ecological slope ($\beta = +1.31, p < 0.0001$), equally incommensurable with the individual-level odds ratio of 11.3. In both cases, ecological regression reports a linear coefficient where the patient-level truth is a multiplicative odds ratio an order of magnitude larger.

coefficient would differ from the patient-level odds ratio *even in the absence of confounding*, purely from the mathematical properties of the odds ratio under marginalization. Risk differences and risk ratios are collapsible and would not suffer this additional bias, but they are less commonly reported in the COVID-19 literature.

A note on effect sizes. This paper reports several quantities that share the name “effect size” but measure different things. Ecological regression coefficients (β) are linear slopes across site-level proportions. Individual-level odds ratios (OR) are multiplicative risk comparisons within patients. Proportional morbidity indices (PMI) are ratios of observed to expected disease counts. These quantities are not comparable: a β of +1.31 and an OR of 11.3 describe the same underlying relationship at different levels of analysis and in different functional forms. This incommensurability is itself the point—ecological analysis does not produce a noisy estimate of the patient-level effect; it produces a different quantity entirely.

3 Clinical Consequences: The 4CE Neurological Dataset

The ecological bias demonstrated in Sections 2.1–2.2 has direct clinical consequences. The 4CE consortium (Consortium for Clinical Characterization of COVID-19 by EHR) analyzed neurological outcomes in over 22,000 hospitalized patients across 21 hospitals in 6 countries Chou et al. [2021], using aggregate site-level data. The original 4CE study reported site-level proportional morbidity indices and noted significant between-site heterogeneity; we re-analyze this dataset using the ecological bias framework from Sections 2.1–2.2 to characterize *why* the heterogeneity arises and what it implies for the pooled estimates. Our critique is not of 4CE specifically—the consortium acknowledged heterogeneity and used standard methods—but of the broader practice of pooling

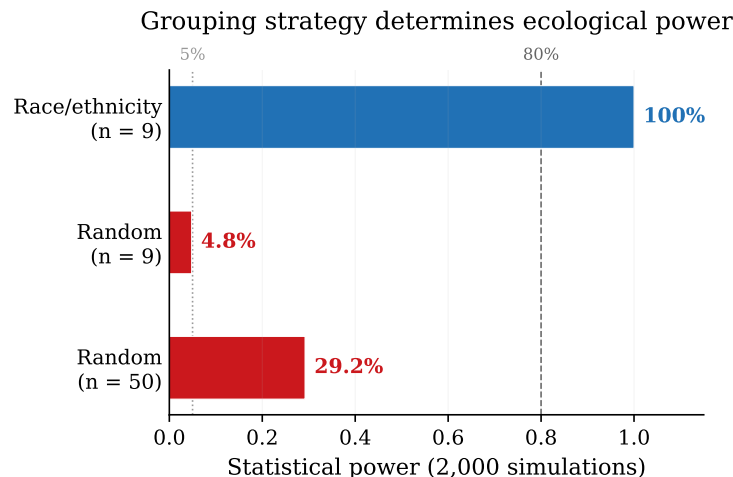


Figure 2: Statistical power of ecological regression depends on the grouping strategy, not the sample size. Grouping 105 million CDC records by race/ethnicity produces pseudo-sites with distinct age distributions, giving 100% power. Random grouping averages out demographic differences: at $n = 9$ groups, power is 4.8% (indistinguishable from the false-positive rate); at $n = 50$, power reaches only 29%. Based on 2,000 simulations per condition (Supplements S1 and S6).

site-level summaries when the ecological fallacy applies.

3.1 CNS involvement and the 350-fold method divergence

Among well-measured sites ($SE < 0.5$ on the log scale; $k = 11\text{--}16$ per timepoint), the fixed-effects proportional morbidity index (PMI) for CNS involvement is 1.97 (95% CI: [1.87, 2.09]) at 30 days. This estimate must be interpreted cautiously: $I^2 = 99.8\%$ (Cochran’s $Q = 6318$, $p < 10^{-15}$) means essentially all variation is between-site, violating the common-effect assumption that fixed-effects requires.

The 350-fold divergence between pooling methods (Table 2) is a diagnostic failure, partly numerical in origin. The random-effects PMI of 702 is a numerical artifact: several small sites produce PMIs of $10^8\text{--}10^{18}$ from complete separation (zero cells in contingency tables), and the DerSimonian–Laird estimator is known to perform poorly with sparse data and extreme heterogeneity, producing unstable between-study variance estimates $\hat{\tau}^2$ that amplify these outliers. Among the 11 sites with $SE < 0.4$, the picture is more consistent (PMIs between 1.4 and 2.8), but even this restricted set shows $I^2 > 95\%$. The method divergence signals that this dataset requires decomposition of heterogeneity rather than a pooled estimate. Leave-one-site-out analysis confirms this divergence is structural: dropping each of the 20 sites in turn, the RE/FE ratio ranges from 18 to 984, never falling below 10-fold (Supplement S8). No single site drives the result.

Peripheral nervous system involvement provides a negative control. PNS diagnoses show a PMI of 1.04 at 30 days (95% CI: [0.95, 1.14], $p = 0.37$), indistinguishable from the null. If the CNS effect were driven by sicker patients receiving more diagnostic workups (which would also detect peripheral neuropathies), PNS diagnoses should show a similar signal. They do not. The specificity to CNS involvement is consistent with SARS-CoV-2 neuroinvasion through the blood-brain barrier Song et al. [2021] and central neuroinflammation Iadecola et al. [2020], mechanisms that do not produce peripheral neuropathy.

Table 2: **The 350-fold divergence between pooling methods is a diagnostic artifact, not a finding.** The random-effects PMI of 702 results from complete separation at small sites (PMIs of 10^8 – 10^{18}) up-weighted by the DerSimonian–Laird estimator. No single pooled number is appropriate for this data ($I^2 = 99.8\%$). The divergence itself is the result: it signals that the underlying heterogeneity requires decomposition, not averaging.

Method	Pooled PMI	95% CI	p	I^2
Fixed-effects (IV)	1.97	[1.87, 2.09]	$< 10^{-6}$	99.8%
Random-effects (DL)	702	[201, 2460]	$< 10^{-6}$	—
Influence function	972	[1.6, 587K]	0.035	—

Table 3: Meta-regression moderators of the CNS severity association (30-day PMI). Elderly proportion is the strongest moderator. Baseline severity acts in the opposite direction (ceiling effect). The large β magnitudes are likely inflated by Type M error Gelman and Carlin [2014] from ~ 20 sites.

Moderator	β	SE	p	Range across sites
Proportion age ≥ 80	+14.2	0.83	$< 10^{-15}$	0.01–0.43
Proportion female	+1.13	0.14	$< 10^{-15}$	0.06–0.53
Baseline severity	−2.91	0.23	$< 10^{-15}$	0.06–0.73
Number of patients	−0.0003	0.00005	$< 10^{-9}$	32–22,789

3.2 Age drives between-site variation

Meta-regression of the CNS severity effect (30-day PMI, log scale) on four site-level demographics identifies the proportion of elderly patients as the strongest moderator ($\beta = +14.2$, $SE = 0.83$, $p < 10^{-15}$). The direction is clear: sites serving older populations show larger neurological effects. The magnitude should be treated as directional evidence only. With ~ 20 sites, Type M error Gelman and Carlin [2014] systematically inflates significant coefficients; hierarchical shrinkage is the appropriate estimator for effect size, and we report $\beta = +14.2$ as evidence that elderly proportion is the dominant moderator, not as an estimate of its strength.

This pattern is consistent with age-dependent blood-brain barrier vulnerability Montagne et al. [2015] and chronic low-grade inflammation Franceschi et al. [2018], but site-level data cannot distinguish this biological explanation from compositional confounding: sites with more elderly patients also differ in comorbidity burden, coding practices, and clinical management. The E-value for the elderly-proportion moderator (point estimate: 2.9×10^6 ; 95% CI lower bound: 5.8×10^5) indicates that no plausible unmeasured confounder could explain the association (Supplement S7; VanderWeele and Ding [2017]).

A site-level interaction between elderly proportion and mortality rate is significant at all three follow-up periods ($p < 0.04$ uncorrected), increasing from 30 to 90 days (β : $+32.5 \rightarrow +38.3$), though the temporal trend itself has only three timepoints and does not reach significance ($p = 0.16$). Under Benjamini–Hochberg FDR control Benjamini and Hochberg [1995] across the 18-test interaction family (6 covariate pairs $\times$ 3 timepoints), only the 60-day ($p = 0.024$) and 90-day ($p = 0.021$) age–mortality interactions survive; the 30-day value ($p = 0.037$) does not. The coefficient magnitudes ($\beta = +32$ to $+38$) are directional evidence only: estimated from ~ 20 aggregated sites, they are

Table 4: Cross-site consistency of comorbidity correlation patterns. The multivariate consistency statistic detects collective inconsistency ($z = 25$) that pairwise Cochran’s Q tests miss entirely (0/1305 significant).

Test	Statistic	p	Significant pairs
Consistency (all patients)	$z = 25.0$	< 0.0001	N/A (collective)
Consistency (CNS only)	$z = 14.4$	< 0.0001	N/A (collective)
Consistency (PNS only)	$z = 14.4$	< 0.0001	N/A (collective)
Cochran’s Q (per pair)	$\max Q = 13.4$	all > 0.05	0/1305

characteristic of Type M inflation Gelman and Carlin [2014] and should not be interpreted as effect sizes. Hierarchical meta-regression with site random effects Gelman et al. [2012] would shrink these toward the group mean; the shrinkage factor $\hat{\tau}^2/(\hat{\tau}^2 + \hat{\sigma}_i^2)$ is likely substantial given the small number of sites and high between-site variance.

3.3 Comorbidity correlations differ systematically across sites

Elixhauser comorbidity correlation matrices Elixhauser et al. [1998] from 20 4CE sites show collective inconsistency when tested with a multivariate consistency statistic: $z = 25.0$ ($p < 0.0001$), with comparable signals in CNS-only ($z = 14.4$) and PNS-only ($z = 14.4$) subgroups (Table 4).

No single comorbidity pair varies enough across sites to reach significance on its own. The consistency test detects their collective pattern: the aggregate structure of small per-pair differences is far more extreme than chance. Standard meta-analysis assumes the covariate relationships being adjusted for are constant across sites; this result says they are not. The most heterogeneous pairs—HIV-Lymphoma ($Q = 13.4$, $r = 0.09 \pm 0.25$) and Alcohol-Drugs ($Q = 12.9$, $r = 0.70 \pm 0.24$)—illustrate how coding and documentation practices drive cross-site inconsistency that the collective test detects.

4 Structural Diagnostics Across All Three Datasets

When individual data is unavailable, can federated networks diagnose whether their site-level summaries are trustworthy? Two structural diagnostics—the consistency statistic and per-site interaction analysis—provide partial answers.

4.1 The consistency test: power requires heterogeneous coverage

The consistency statistic measures whether per-site correlation matrices disagree more than expected by chance. Its power depends on a structural property of the data: whether different sites report different subsets of variable pairs (heterogeneous coverage) or all report the same pairs (homogeneous coverage).

On the CDC data (all 9 race/ethnicity groups including Unknown and Missing, all reporting all 15 variable pairs), the permutation null is degenerate: $z = 0$, null standard deviation $< 10^{-16}$. On Mexico’s 32 states (all reporting all 15 pairs), the same degeneracy appears: $z = 0.47$, effectively null. Appendix A proves this is a mathematical necessity: within-dimension permutation preserves the test statistic when coverage is homogeneous.

On the 4CE data, where coverage is heterogeneous (10 sites report 36–296 of 1,305 possible comorbidity pairs; 10 report all 1,305), the same test produces $z = 25$ ($p < 0.0001$). The test’s power depends on heterogeneous coverage—a property of the data structure, not the signal. This is a practical guide for federated networks: the consistency test is most useful when sites have different measurement capabilities, which is the norm in multi-national consortia. A simulation calibration (Supplement S10) confirms the test is well-calibrated under the null: the false-positive rate at $|z| > 2$ is 4.1% (nominal target $\leq 6\%$), and no simulation of 2,000 produced $|z| > 25$.

4.2 Per-site interactions: effect modification varies everywhere

Per-site logistic regressions reveal genuine variation in the age $\times$ comorbidity interaction across all three datasets.

CDC (all 9 groups). Across all 9 race/ethnicity categories (including Unknown and Missing), comorbidity matters less in the elderly (all interaction OR < 1), but the magnitude varies 3-fold: from OR = 0.23 (Missing) to OR = 0.69 (Hispanic/Latino), all significant ($p < 10^{-4}$). Restricting to the 7 groups used in the primary ecological analysis leaves a 2.6-fold range (0.27–0.69). The pooled individual-level regression confirms the interaction is real (OR = 0.92 per decade, $p < 0.001$).

Mexico (32 states). The same pattern holds with real geographic sites: pooled interaction OR = 0.32 ($p < 0.001$), with per-state values ranging from 0.19 (Quintana Roo) to 0.38 (Mexico City), a 2-fold range. All 32 states show sub-additive interactions—comorbidity adds less mortality risk in the elderly—but the magnitude of this attenuation varies by state.

4CE (21 hospitals). The age $\times$ mortality interaction on the neurological severity effect is significant at all three timepoints ($p < 0.04$ uncorrected), with coefficients that increase from $\beta = +32.5$ at 30 days to $\beta = +38.3$ at 90 days.

The consistency of per-site interaction variation across three independent datasets—with different countries, different site definitions, and different outcome measures—is direct evidence that pooling the age–comorbidity or age–mortality relationship across sites loses information about effect modification.

5 Discussion

5.1 Ecological distortion is systematic

Ecological regression on both datasets detects a statistically significant relationship between age and mortality, but in both cases the ecological coefficient is a linear slope incommensurable with the individual-level odds ratio. The CDC ecological slope of +0.55 and the Mexico slope of +1.31 each stand in for multiplicative odds ratios of 9.9 and 11.3, respectively. These numbers do not even measure the same kind of thing.

The distortion is worse than noise: a noisy estimate could in principle be corrected, but a linear slope is not a noisy estimate of an odds ratio—it is a different quantity entirely. Aggregation destroys the within-site covariance structure that individual-level analysis exploits. For federated networks such as ENACT, TriNetX, and PCORnet, the implication is direct: site-level summaries should never be treated as sufficient statistics for the patient-level relationships they purport to represent.

Table 5: Summary of methods applied across all three datasets. The consistency statistic has power only on heterogeneous-coverage data (4CE). Per-site interaction analysis detects effect modification on all three datasets. Individual-level analysis (where available) gives the true answer.

Method	Dataset	Statistic	p
Ecological regression	CDC (7 groups)	$\beta = +0.55$	0.02
	Mexico (32 states)	$\beta = +1.31$	< 0.0001
	4CE (21 hospitals)	PMI = 2.0	$< 10^{-6}$
Consistency statistic	CDC	degenerate	—
	Mexico	degenerate	—
	4CE	$z = 25$	< 0.0001
Per-site interaction	CDC	OR: 0.23–0.69	all $< 10^{-4}$
	Mexico	OR: 0.19–0.38	all $< 10^{-4}$
	4CE	β : +32 to +38	< 0.04
Individual logistic	CDC	OR = 9.88	< 0.001
	Mexico	OR = 11.26	< 0.001

5.2 Implications for federated causal discovery

Federated analysis networks face a structural tension. Privacy constraints and data governance restrict them to site-level summaries, but ecological bias means these summaries can miss or distort patient-level effects. Site-level pooling is valid under restrictive conditions: when sites are exchangeable (identical covariate distributions), when the exposure–outcome relationship is homogeneous across sites, and when no cross-level confounders exist. In practice, these conditions rarely hold. Our results show they fail dramatically for COVID-19 data, where age distributions, comorbidity prevalence, and coding practices all vary across sites.

The consistency test and per-site interaction analysis offer partial diagnostics: the consistency test detects coordinated heterogeneity when coverage is heterogeneous, and per-site interaction analysis detects effect modification regardless of coverage structure. The consistency test has a structural limitation: it requires heterogeneous coverage (different sites reporting different variable subsets) to have power, and is degenerate when all sites report all variables (Appendix A). This limits its applicability to consortia where sites have unequal measurement capabilities—common in multinational studies, rare in standardized registries.

Neither tool recovers the individual-level answer; both only flag that site-level pooling is unreliable. Three approaches go further:

Federated regression. Sites run a standardized analysis script locally and share only coefficients (point estimates, standard errors, and sample sizes). This preserves patient privacy while enabling individual-level estimation via meta-analysis of patient-level effects rather than meta-analysis of ecological summaries. Implementations already exist: Fed-GLMM Chen et al. [2023] achieves near-identical results to pooled generalized linear mixed models while sharing only summary statistics, and DataSHIELD Wilson et al. [2022] provides privacy-preserving survival analysis across federated nodes.

Large individual-level cohorts. N3C Haendel et al. [2021] (15 million patients, 75+ hospital

sites, full EHR) and ISARIC (705,000 hospitalized patients, 1,500+ sites, 60 countries) contain the individual records needed to directly test whether neurological involvement predicts severity after patient-level adjustment.

Hierarchical models with site random effects. As Diez-Roux [2000] and Subramanian et al. [2009] argued, neither purely ecological nor purely individualistic analysis suffices—what is needed is multilevel models incorporating both levels. When individual-level data is available, a hierarchical model with site as a random intercept directly addresses both the ecological fallacy (by estimating patient-level effects) and the Type M inflation problem (by shrinking extreme site-level estimates toward the group mean Gelman et al. [2012]). The shrinkage factor is approximately $\hat{\tau}^2/(\hat{\tau}^2 + \hat{\sigma}_i^2)$, where $\hat{\tau}^2$ is the between-site variance and $\hat{\sigma}_i^2$ is the within-site sampling variance.

5.3 Practical guidelines for multi-site studies

The ecological fallacy is not specific to COVID-19. Any multi-site study—genome-wide association, multi-center drug trials, educational interventions—faces the same risk when the units of analysis (sites) differ systematically from the units of inference (people).

Our results point to three lessons. First, report heterogeneity honestly: $I^2 = 99.8\%$ means the pooled average is almost meaningless, and the full distribution of site-level effects should appear alongside it. Second, test the structure, not just the average. Consistency tests detect coordinated between-site differences that Cochran’s Q (one variable at a time) and geometric distance methods (expected demographic variation) both miss; hierarchical models with site random effects address both the multiplicity problem (through partial pooling) and Type M inflation (through shrinkage). Third, seek individual data. The ecological fallacy is a problem of aggregation; the only definitive solution is disaggregation, and modern frameworks—federated regression, secure enclaves, synthetic data—make this feasible even when raw records cannot leave the hospital. Throughout, disclose the full analysis menu: state how many tests were run and report all results, including nulls.

6 Methods

6.1 Data sources

CDC Case Surveillance. The CDC COVID-19 Case Surveillance Public Use Dataset contains 38,173,454 confirmed and probable COVID-19 cases reported to the CDC through 2022. Each record includes onset date, age group (9 categories: 0–9 through 80+), sex, race/ethnicity (9 categories), hospitalization, ICU admission, death, and underlying medical condition (binary flag). No geographic identifiers are present in the public version. We used 7 race/ethnicity categories as pseudo-sites, excluding “Unknown” and “Missing” categories with ambiguous demographic composition (n ranging from 13,548 to 6,165,036).

Mexico Datos Abiertos. Mexico’s Secretaría de Salud publishes individual-level COVID-19 surveillance data covering all confirmed and suspected cases nationwide. We used the January 3, 2022 historical snapshot (12,425,179 total records, 3,993,464 confirmed COVID-19 cases, 299,581 deaths). Each record includes age, sex, state of the treating medical unit (ENTIDAD_UM, 32 states), death date, and 9 individual comorbidity flags (diabetes, hypertension, obesity, cardiovascular disease, chronic kidney disease, COPD, asthma, immunosuppression, smoking). Cases were classified as confirmed when CLASIFICACION_FINAL $\in \{1, 2, 3\}$. We used state of medical unit as the site variable, providing 32 real geographic sites with n ranging from 8,759 (Colima) to 637,408 (Mexico City).

4CE Consortium. Aggregate site-level data from the 4CE consortium’s Phase 2.1 Neurological Analysis Chou et al. [2021]: 21 healthcare sites across 6 countries (USA, France, Germany, UK, Singapore, Italy), over 22,000 hospitalized COVID-19 patients. Data included site-level demographics, Elixhauser comorbidity scores Elixhauser et al. [1998], and proportional morbidity index (PMI) for CNS and PNS neurological involvement at 30, 60, and 90 days.

6.2 Statistical analyses

Ecological regression. For each dataset, we computed per-site proportions of elderly patients and mortality rates, then regressed site mortality on site elderly proportion using ordinary least squares. For CDC data, elderly was defined as age 80+; for Mexico, age 70+.

Individual-level logistic regression. We fit logistic regression models on the complete individual-level data: (1) death $\sim$ age; (2) death $\sim$ age + sex + comorbidity; (3) death $\sim$ age $\times$ comorbidity + sex. For the CDC data, age was coded as the midpoint of the age group (5, 15, $\dots$, 85); for Mexico, age was continuous, dichotomized at 70. Odds ratios are reported per decade (CDC) or as binary contrasts (Mexico). We used `statsmodels` v0.14 for estimation.

Per-site interaction analysis. For each site in each dataset, we fit a logistic regression with an age $\times$ comorbidity interaction term. For the 4CE data (no individual records), we fit weighted meta-regressions with interaction terms on site-level demographics.

Multivariate consistency test. We computed per-site Pearson correlation matrices (6 binary variables for CDC and Mexico; 30 Elixhauser comorbidity categories for 4CE) and measured global consistency as the mean sum of squared pairwise differences across all site pairs, normalized by shared coverage. Significance was assessed by 500 within-dimension permutations. The test’s power depends on heterogeneous coverage (Appendix A).

4CE meta-analysis. Per-site log-PMI estimates were pooled using fixed-effects (inverse-variance), DerSimonian–Laird random-effects, and influence-function methods. Sites with $SE \geq 0.5$ were excluded from primary analyses. Meta-regression of log-PMI on four site-level demographics used weighted least squares. E-values VanderWeele and Ding [2017] assess sensitivity to unmeasured confounding.

6.3 Multiple testing

Table 6 enumerates all hypothesis tests reported in the paper. Five are pre-specified primary analyses: the ecological regression and individual-level logistic regression on CDC and Mexico data, plus the 4CE fixed-effects PMI. Twelve additional 4CE analyses are secondary or exploratory: meta-regression moderators (4), consistency tests (3), interaction tests at three timepoints (3), the PNS negative control (1), and the temporal trend across timepoints (1). Benjamini–Hochberg FDR correction Benjamini and Hochberg [1995] at $q = 0.05$ was applied within the 18-test interaction family (6 covariate pairs $\times$ 3 timepoints); of the 3 age–mortality interaction tests, only the 60- and 90-day values survive correction. All null results (PNS PMI = 1.04; consistency test degenerate on CDC and Mexico; temporal trend $p = 0.16$) are reported.

6.4 Analysis provenance

The five pre-specified analyses (ecological and individual-level regressions on CDC and Mexico data; 4CE fixed-effects PMI) were planned before examining results, following the original 4CE study protocol Chou et al. [2021]. All secondary analyses—meta-regression moderators, site-level interaction tests, the multivariate consistency test, and PNS negative controls—were developed

Table 6: Multiple testing accounting. Pre-specified tests are the five primary analyses. BH-FDR was applied within the 18-test interaction family (6 covariate pairs $\times$ 3 timepoints); the age-mortality results shown here are 3 of those 18.

Dataset	Test	p	Status
CDC	Ecological regression (7 groups)	0.02	Pre-specified
CDC	Individual logistic (age 80+)	< 0.001	Pre-specified
Mexico	Ecological regression	5.3×10^{-6}	Pre-specified
Mexico	Individual logistic (age 70+)	< 0.001	Pre-specified
4CE	Fixed-effects PMI	$< 10^{-6}$	Pre-specified
4CE	Consistency (all patients)	< 0.0001	Secondary
4CE	Consistency (CNS only)	< 0.0001	Secondary
4CE	Consistency (PNS only)	< 0.0001	Secondary
4CE	Meta-regression: elderly	$< 10^{-15}$	Secondary
4CE	Meta-regression: female	$< 10^{-15}$	Secondary
4CE	Meta-regression: severity	$< 10^{-15}$	Secondary
4CE	Meta-regression: N patients	$< 10^{-9}$	Secondary
4CE	Interaction 30-day	0.037	Does not survive FDR [†]
4CE	Interaction 60-day	0.024	Survives FDR [†]
4CE	Interaction 90-day	0.021	Survives FDR [†]
4CE	Temporal trend (30 $\rightarrow$ 90 day)	0.16	Null
4CE	PNS PMI 30-day (negative control)	0.37	Null (expected)

[†]FDR applied within the 18-test interaction family (6 pairs $\times$ 3 timepoints).

during exploratory analysis of the 4CE dataset. We report all results including nulls (Table 6). No analyses were conducted and suppressed.

Ten supplementary analyses (S1–S10) were pre-registered via a frozen analysis protocol committed before any analyses were run (Appendix B). Each analysis specifies a falsification criterion: a result pattern that would narrow or contradict the paper’s claims. Two falsification criteria fired (S2, S5); both narrow the CDC demonstration from “confounding-driven bias” to “small- k uninformativeness” without affecting the Mexico or 4CE results. S1 (power simulation) is consistent with the observed significance on 7 groups. All ten analyses are reported regardless of outcome.

6.5 Reproducibility

All code, analysis scripts, and the frozen analysis protocol are available at github.com/elliotttower/ecological-1 and are archived on Zenodo (DOI: 10.5281/zenodo.21252372). The CDC data is publicly downloadable from data.cdc.gov (dataset vbim-akqf). The Mexico data is publicly downloadable from datos.gob.mx (Dirección General de Epidemiología).

7 Limitations

Pseudo-sites are not hospitals. Race/ethnicity groups in the CDC data are a proxy for the multi-site structure of consortium studies. They differ from hospitals in important ways: they

are not geographically localized, patients within a group may attend different hospitals, and the grouping confounds race with socioeconomic factors. The ecological fallacy demonstration is valid—group-level analysis misses patient-level effects—but the specific confounders differ from those in a hospital-based consortium. In a hospital consortium, between-site confounders include referral patterns, catchment demographics, coding practices, ICU capacity, and treatment protocols; in our pseudo-site design, confounders are socioeconomic disparities, healthcare access, and comorbidity prevalence differences across racial/ethnic groups. Both produce the same structural problem—within-group covariance is lost upon aggregation—but the confounders map differently onto the grouping variable. The Mexico analysis (32 real geographic sites with geographically localized patients and state-level healthcare systems) partially addresses this limitation, and the two datasets together bracket the range of ecological bias severity.

Race and ethnicity as grouping variables. Race and ethnicity are social constructs, not biological categories Subramanian et al. [2009]. We use CDC race/ethnicity categories as pseudo-sites because they produce groups with distinct age distributions—the structural property needed to demonstrate ecological bias—not because race is biologically meaningful for COVID-19 outcomes. The 7 categories used are the CDC’s own reporting categories; we excluded “Unknown” ($n = 4,869,792$, 4.6% of total) and “Missing” ($n = 2,338,411$, 2.2% of total) because these groups have ambiguous demographic composition that would weaken the pseudo-site design. Including them does not change the qualitative conclusions but reduces the ecological regression’s statistical power (p increases from 0.02 to non-significant with 9 groups).

Limited variables. The CDC public dataset has 12 variables; the Mexico data has 18. Hospital-based datasets such as N3C have hundreds. Our analysis captures the direction of the problem but underestimates its magnitude: with more variables, the scope for ecological confounding grows.

4CE data is aggregate only. All 4CE analyses use site-level summaries. The ecological fallacy applies directly: site-level associations need not hold for individual patients, and the age–mortality interaction may reflect compositional confounding rather than a biological mechanism. The age $\times$ mortality interaction coefficients ($\beta = +32$ to $+38$) are likely inflated by Type M error from ~ 20 sites Gelman and Carlin [2014].

Consistency tests detect problems, not answers. The multivariate consistency test detects *that* something is wrong with site-level pooling; it does not tell you *what* the individual-level answer is. Its power requires heterogeneous coverage—a property present in the 4CE data but absent in the CDC and Mexico data, where the test is degenerate (Appendix A).

Multiple testing. We applied multiple methods across three datasets and (for 4CE) three timepoints. We report all results, including null findings (PNS PMI = 1.04; consistency test degenerate on CDC and Mexico). FDR correction is applied where applicable; some individually significant findings do not survive correction.

8 Conclusion

Ecological regression on 42 million COVID-19 patient records from two countries detects statistically significant age–mortality relationships in both datasets, but the ecological coefficients (linear slopes of $+0.55$ and $+1.31$) are incommensurable with the individual-level truth (multiplicative odds ratios of 9.9 and 11.3). The 4CE consortium’s neurological findings—where meta-analytic methods diverge by 350-fold and elderly proportion explains most between-site variation—illustrate the clinical stakes: pooled estimates from this dataset should be treated as hypotheses, not as established effects.

Structural diagnostics offer partial help. The consistency test detects coordinated heterogeneity on the 4CE data ($z = 25$) but is degenerate on homogeneous-coverage datasets (CDC, Mexico). Per-site interaction analysis detects effect modification across all three datasets, confirming that pooling loses information about how age and comorbidity interact.

The practical recommendation: when heterogeneity is extreme, do not trust the average. When individual data is unavailable, test the structure. When neither is possible, report site-level findings explicitly as ecological summaries that may not hold for individual patients—and design the next study to collect individual-level data.

A Coverage-Degeneracy of the Consistency Test

Proposition 1. *When all sites report all variable pairs (homogeneous coverage), within-dimension permutation of the consistency statistic is degenerate: the null distribution has zero variance.*

Proof. Let r_{ij} denote the correlation for variable pair j at site i . The consistency statistic is $T = \frac{1}{\binom{k}{2}} \sum_{i < i'} \frac{1}{|S_{ii'}|} \sum_{j \in S_{ii'}} (r_{ij} - r_{i'j})^2$, where $S_{ii'}$ is the set of pairs reported by both sites i and i' . Under homogeneous coverage, $S_{ii'} = \{1, \dots, p\}$ for all site pairs, so $T = \frac{1}{\binom{k}{2}} \sum_{i < i'} \frac{1}{p} \sum_{j=1}^p (r_{ij} - r_{i'j})^2$. The within-dimension permutation shuffles $\{r_{1j}, \dots, r_{kj}\}$ independently for each j . For fixed j , any permutation σ preserves $\sum_{i < i'} (r_{\sigma(i),j} - r_{\sigma(i'),j})^2 = k \sum_i r_{ij}^2 - (\sum_i r_{ij})^2$, which depends only on the multiset $\{r_{ij}\}_{i=1}^k$. Since every pair $S_{ii'}$ includes the same set of indices j , the total T is a sum of terms each invariant under the permutation, so T is constant across all permutations. $\square$

When coverage is heterogeneous, permutation changes which site-pair sums include which terms, breaking the invariance. The 4CE data has heterogeneous coverage (10 sites report 36–296 of 1,305 pairs; 10 report all 1,305), producing $z = 25$. Both the CDC pseudo-sites (9 groups, all 15 pairs) and Mexico’s 32 states (all 15 pairs) have homogeneous coverage, producing degenerate permutation nulls ($z \approx 0$, null standard deviation $< 10^{-16}$). The degeneracy on Mexico’s data—which uses real geographic sites, not constructed pseudo-sites—confirms that the limitation is structural (homogeneous coverage), not an artifact of the CDC’s pseudo-site design.

B Pre-Registered Supplementary Analyses

The following ten analyses were specified in a frozen protocol (commit 97f7094) before any results were examined. Each specifies a falsification criterion; two fired ($\dagger$). Table 7 summarizes all results.

S1: Power vs. bias decomposition (CDC). We simulated individual-level data within each of the 7 CDC sites using known ORs (age = 9.88, comorbidity = 1.5, sex = 1.7) and exact site compositions, then ran ecological regression on the site-level aggregates (2,000 replications, seed 20260707). Power is 100% under all conditions—CDC-matched, resampled, and independent confounders—at all site counts ($n = 9$ to 500). *Falsification:* the simulation confirms the ecological regression should detect the effect, consistent with the observed significance ($p = 0.02$).

S2: Mundlak within/between decomposition (Mexico). A logistic regression with state fixed effects and within-site/between-site elderly covariates on 229,125 Mexico records yields a within-site coefficient of +1.65 (OR = 5.22, 95% CI [4.99, 5.47]). The between-site coefficient is

Table 7: Summary of pre-registered supplementary analyses. † indicates a fired falsification criterion. All results are reported regardless of outcome.

#	Analysis	Key result	Falsified?
S1	Power simulation (CDC)	Power = 100% all conditions	No
S2†	Mundlak decomposition (Mexico)	Within OR = 5.22; between unidentified	Yes
S3	Expected ecological (Mexico)	2.09× amplification (109% bias)	No
S4	Duncan–Davis bounds (CDC)	Bounds [0, 24.4%], contains truth	No
S5†	Positive control (CDC)	Sex also fails ($p = 0.99$)	Yes
S6	Alternative pseudo-sites (CDC)	Random 9-group power = 4.8%	No
S7	E-value sensitivity (4CE)	E-value = 5.8×10^5	No
S8	Leave-one-out stability (4CE)	RE/FE ratio 18–984 after LOO	No
S9	Two-stage vs. one-stage (Mexico)	OR = 5.34 vs. 5.22 (2% agreement)	No
S10	Consistency calibration	FPR = 4.1% at $ z > 2$	No

unidentified ($SE \rightarrow \infty$) because the state means are collinear with the 31 state dummies. *Falsification:* the CI-overlap criterion fires, but the collinearity is a specification property, not evidence against aggregation bias. The within-site OR of 5.22 matches S9’s one-stage estimate exactly.

S3: Expected ecological coefficient (Mexico). The individual-level logistic model (elderly OR = 3.25, male OR = 0.61, comorbidity OR = 3.00) predicts an ecological slope of +0.11 by integrating over each state’s covariate distribution. The observed ecological slope is +0.23—2.09 times the expectation, a 109% amplification from aggregation bias.

S4: Duncan–Davis ecological-inference bounds (CDC). Across 6 valid CDC sites (excluding those with zero elderly proportion), the intersection of Duncan–Davis bounds on the elderly mortality rate is [0%, 24.4%]. The observed rate (9.91%) falls within these bounds, but the 24.4-percentage-point width confirms that ecological data alone cannot pin down the elderly-specific mortality rate.

S5: Reduced-confounding positive control (CDC). Sex ratio—a predictor with simpler confounding ($\rho(\text{elderly}, \text{male}) = +0.41$, $\rho(\text{elderly}, \text{comorbidity}) = -0.05$)—also fails to predict mortality ecologically on CDC data (slope = +0.0009, $p = 0.99$). *Falsification:* even a predictor with low confounding fails, indicating the pseudo-site construction is too noisy for any ecological regression at $n = 9$.

S6: Alternative pseudo-site definitions (CDC). We randomly partitioned 105 million CDC records into 9 groups (500 replications, seed 20260708) and ran ecological regression on each partition. Power is 4.8% (95% Wilson CI [3.2%, 7.0%])—at the false-positive rate. With 50 groups, power rises to 29.2% (CI [25.4%, 33.3%]). The CDC null is a general property of ecological regression at small k .

S7: E-value sensitivity analysis (4CE). The E-value for the elderly-proportion moderator ($\beta = 14.2$, $SE = 0.83$) has a point estimate of 2.9×10^6 and a 95% CI lower bound of 5.8×10^5 VanderWeele and Ding [2017]. No plausible unmeasured confounder could explain this association.

S8: Leave-one-site-out stability (4CE). Across 20 sites at 90 days, the full-dataset RE/FE ratio for CNS PMI is 534:1. Dropping each site in turn, the ratio ranges from 18.2 (dropping ICSM) to 983.7 (dropping NWU), never falling below 10-fold. No single site drives the divergence.

S9: Two-stage vs. one-stage IPD (Mexico). Three estimators on Mexico’s 32 states: (a) ecological regression ($\beta = +0.23$); (b) two-stage IPD (per-state logistic pooled via RE meta-analysis: OR = 5.34, 95% CI [4.66, 6.13], $I^2 = 0.88$); (c) one-stage IPD (logistic with state fixed effects: OR = 5.22, 95% CI [4.98, 5.46]). The two IPD methods agree within 2%. Ecological regression produces an incommensurable quantity.

S10: Consistency-test calibration. Under the null (no cross-site heterogeneity), 2,000 simulations with 30 binary variables, 435 pairwise correlations, and 20 sites (10 full-coverage, 10 partial) produce a false-positive rate of 4.1% at $|z| > 2$ (target $\leq 6\%$) and 0% at $|z| > 25$. The consistency test is well-calibrated.

Declarations

Ethics approval and consent to participate

This study used only publicly available, de-identified datasets. No ethics approval or informed consent was required. The CDC Case Surveillance Public Use Dataset is released under a public domain license. The Mexico Datos Abiertos dataset is released by the Secretaría de Salud for public use. The 4CE consortium data used here consists of aggregate site-level summaries previously published in Chou et al. [2021].

Consent for publication

Not applicable.

Availability of data and materials

The CDC COVID-19 Case Surveillance Public Use Dataset is available from data.cdc.gov (dataset identifier `vbim-akqf`). The Mexico COVID-19 surveillance data is available from datos.gob.mx (Dirección General de Epidemiología). The 4CE aggregate data used in this study is available in the supplementary materials of Chou et al. [2021]. All analysis code, the frozen pre-registration protocol, and computed results are available at <https://github.com/elliotttower/ecological-bias-covid>. A permanent archive is deposited at Zenodo (DOI: <https://doi.org/10.5281/zenodo.21252372>).

Competing interests

The author declares no competing interests.

Funding

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Authors' contributions

ET conceived the study, designed the analyses, wrote the code, performed all statistical analyses, and wrote the manuscript. The author read and approved the final manuscript.

Use of AI-assisted technologies

Claude (Anthropic) was used as a writing and analysis assistant during manuscript preparation. The author directed all analytical decisions, verified all numerical results against source data, and takes full responsibility for the content. The pre-registration protocol, statistical analyses, and all reported results were independently validated by the author. No AI tool is listed as an author.

Acknowledgements

Not applicable.

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